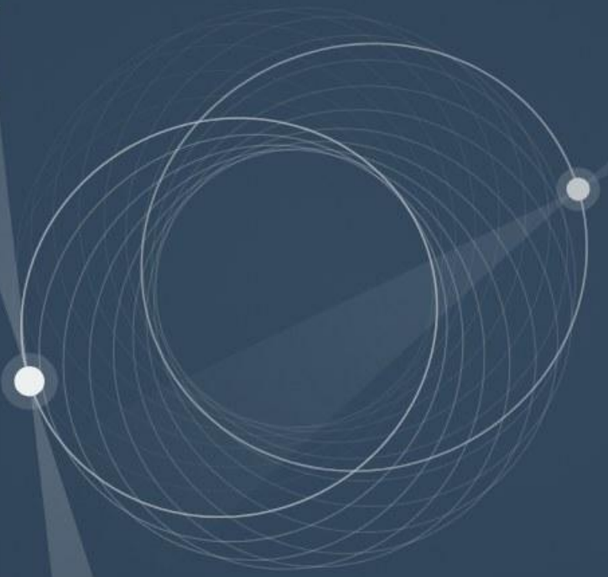
PSR J0737−3039A · BAYESIAN INFERENCE

The moment of inertia of *pulsar A*

Bayesian predictions for the moment of inertia of PSR J0737−3039A and tests of the strange-star hypothesis



Rafael C. R. de Lima

with João Vitor Zastrow and Jonas P. Pereira

UDESC/CCT · Joinville · UnB · Brasília

OUTLINE

Four parts

I **The double pulsar**
What timing measures and why I_A carries the structure of the star

II **Two families**
Self-bound strange matter and nucleonic polytropes, with the same solvers

III **The inference**
Public data, likelihoods, samplers and a biased move

IV **Results**
Predictions for pulsar A, evidences and what a future measurement adds

THE QUESTION

From compatibility *to probability*

PREVIOUS PAPER

Could it be compatible?

A scan of strange-star equations of state, fixed cuts in maximum mass, radius and deformability, and three nucleonic reference EOSs.

The retained models satisfy the timing bound on I_A. No probability is assigned to any of them.

THIS WORK

How probable, and under which choices?

Posterior of the strange family from public data, predictive distributions of I_A, R_A and Λ_A, and a flexible nucleonic family under the same data.

Then: how much a future measurement of I_A would add.

THE RESULT, BEFORE THE PATH

The answer in *three numbers*

1.44

I_A of strange stars, in 10^{45} g cm^2. 90% interval: 1.29–1.58

+0.21

ln $\mathcal{B}$ strange/nucleonic. Between −1.5 and +1.5 across prior and data variants

1.5 km

How much smaller strange stars are than nucleonic stars at the same I_A

> Current data **do not discriminate** between these two families. I_A alone separates little in the simulated cases; the radius acts in the direction in which they differ.

I

PART I

The double pulsar

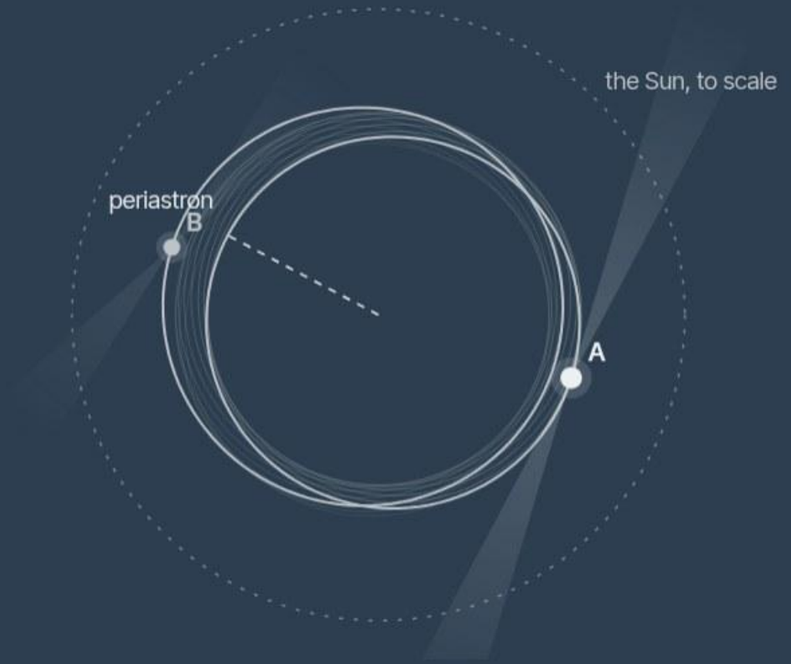
A 23 ms clock in a two-and-a-half-hour orbit, with a known mass

PSR J0737-3039

Two pulsars in *2.45 hours*

The orbit to scale, with the periastron turning

• ANIMATED



Years of timing 16.2 yr

Periastron turned 273.8° (273.8° in this turn) Lense–Thirring part −0.0088°

Orbital period	0.10225 d
Eccentricity	0.0878
Spin of A	22.7 ms
Spin of B	2.8 s
Periastron advance	16.899323(13) °/yr
Timing baseline	16.2 yr

The periastron completes a turn in about 21 years. The whole orbit would fit inside the Sun.

Kramer et al. 2021, PRX 11, 041050, Table 4 · Burgay et al. 2003 · Lyne et al. 2004

The mass is fixed; *the structure is missing*

MAIN SOLUTION OF KRAMER ET AL.

$$m_A = 1.338185^{+0.000012}_{-0.000014}\ M_\odot$$

Periastron advance and Shapiro-delay shape, with I_A estimated from radius limits of another EOS analysis

SOLUTION WITH NO EOS ASSUMPTION: THE ONE WE USE

$$m_A = 1.338183^{+0.000078}_{-0.000052}\ M_\odot$$

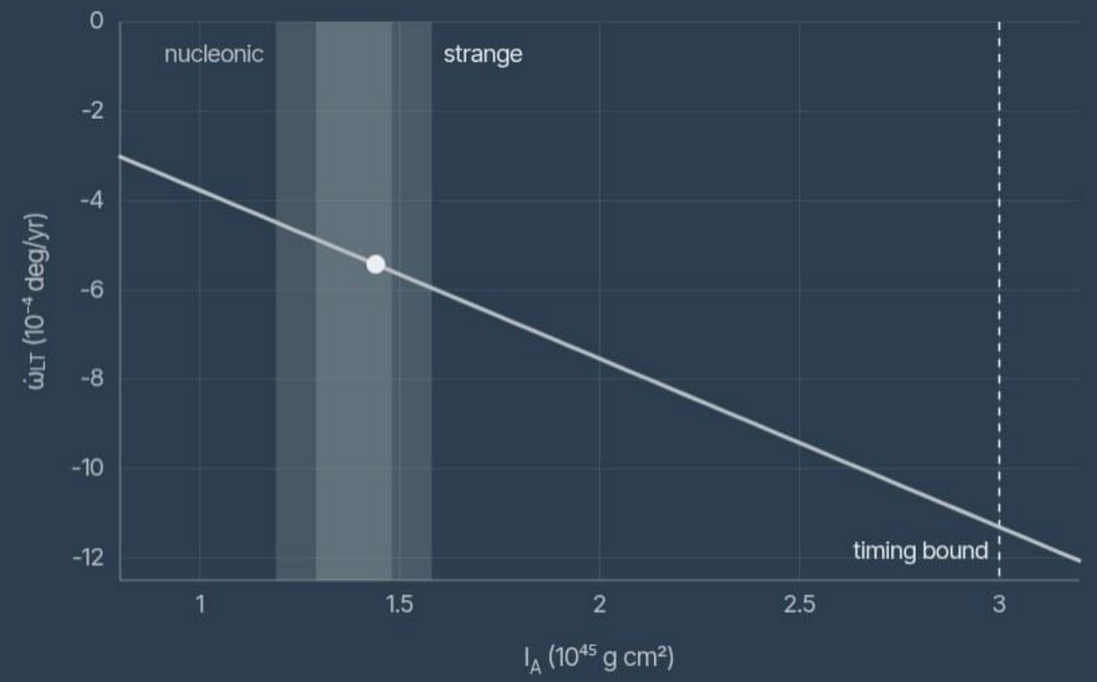
Periastron advance, Shapiro-delay shape and orbital decay together, with I_A free

> Between the two masses, I_A changes by less than $\mathbf{10^{-4}}$ in relative terms. We use the second because it does not assume what we test.

Kramer et al. 2021, §VI.2.1 and §VI.2.3 · Manuscript · §IV

The spin of A *drags the periastron*

Lense–Thirring term against I_A • INTERACTIVE



I_A (10^{45} g cm²) 1.44

$\dot{\omega}_{LT}$ −5.43×10⁻⁴ °/yr Share of $\dot{\omega}$ 3.2×10⁻⁵ Error on $\dot{\omega}$ ±0.13×10⁻⁴

CONTRIBUTION TO $\dot{\Omega}$	°/YR
1PN	16.898703
2PN	+0.000439
Lense–Thirring	−0.000377 I_{45}

PN means post-Newtonian: a series in $(v/c)^2$. 1PN is order v^2/c^2, and 2PN is order v^4/c^4. Here $v/c \approx 2\times10^{-3}$.

$\dot{\omega}_{LT}$ is the part of $\dot{\omega}$, the periastron advance rate, caused by the spin of A, which drags spacetime. It reduces the advance and is proportional to I_A, with $I_{45} = I_A / 10^{45}$ g cm².

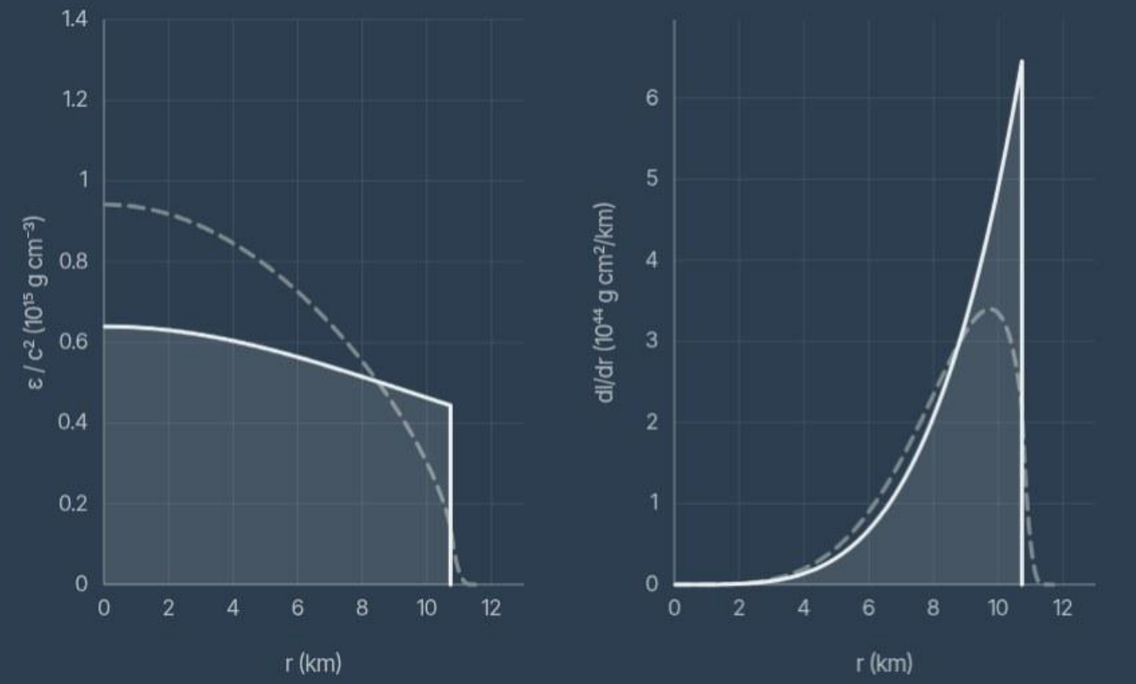
Current bound: $\mathbf{I_A < 3.0\times10^{45}\ g\ cm^2}$ (90%). Its distribution is not public, so it does not enter as a likelihood. Published forecasts anticipate about 10% by 2030.

Hu et al. 2020, Table 1 (2006 masses) · Kramer et al. 2021, eq. 17 · Hu & Freire 2024

I measures how the mass *is distributed*

Two stars of m_A = 1.338183 M⊙, computed here

• LIVE TOV + HARTLE



λ −3.4

$B^{1/4}$ (MeV) 161.5

Strange 10.75 km 1.434 · SLy 11.74 km 1.285 ×10⁴⁵ g cm²

Hartle 1967 · Manuscript · §III and §VIII · in this panel, first-order TOV + Hartle in the browser

$$I = \frac{8\pi}{3} \int \frac{r^4 (\varepsilon + p)\, e^{-\nu/2}}{\sqrt{(1 - 2m/r)}} \frac{\bar{\omega}}{\Omega}\, dr$$

With the mass given, I_A weighs the outer layers.

The strange star is nearly uniform and ends at finite density. For the same radius, that gives a larger moment of inertia.

— strange (sliders) - - SLy, piecewise polytropes

II

PART II

Two families

Self-bound quark matter against a flexible nucleonic parameterization

SELF-BOUND

The *strange-matter* hypothesis

$$\Omega = \frac{3}{4\pi^2}\left(-a_4\,\mu^4 + a_2\,\mu^2\right) + B$$

Grand potential of quark matter · μ is the quark chemical potential

$a_4 \le 1$	perturbative QCD corrections
a_2	strange-quark mass and pairing
B	effective bag constant

> If three-flavor matter is the ground state, a whole star can be made of it: self-bound, with **finite density at the surface**. We model a bare surface.

Alford et al. 2005 · Manuscript · §II

Stable with three flavors, *unstable with two*

$$B_{max} = g_0^2(g_0^2 a_4 - 9a_2) / 108\pi^2$$

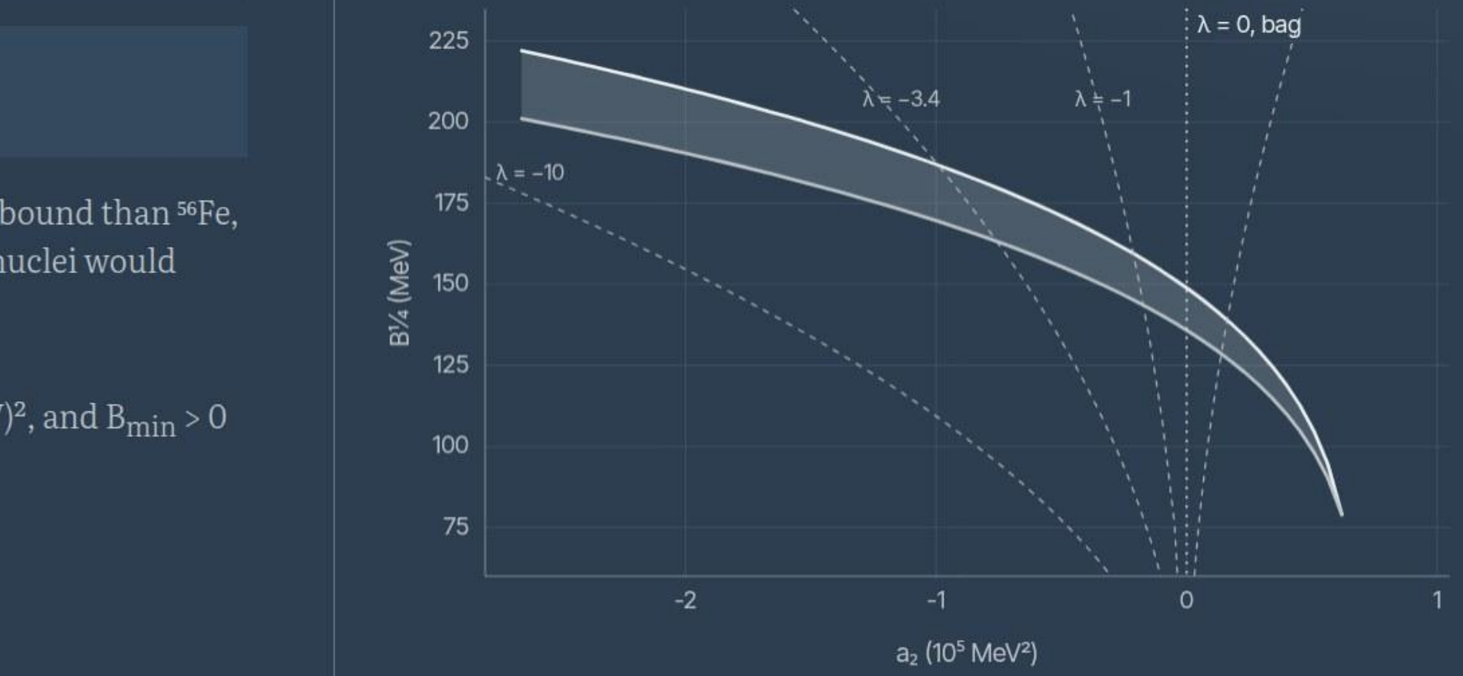
$$B_{min} = \frac{g_0^2}{54\pi^2}\left[\frac{4g_0^2 a_4}{(1+2^{1/3})^3} - 3a_2\right]$$

$g_0 = 930$ MeV. Three-flavor matter must be more bound than ^{56}Fe, and two-flavor matter less: otherwise ordinary nuclei would convert.

The window exists for $a_2 \le \alpha\, a_4$, with $\alpha \approx (297\ \text{MeV})^2$, and $B_{min} > 0$ throughout it.

Prior domain in (a_2, $B^{1/4}$)

• INTERACTIVE



a_4 0.70

At $a_2 = 0$: $B^{1/4}$ from 135 to 148 MeV · a_2 up to 0.62×10^5 MeV²

Zastrow et al. 2025 · Manuscript · §II and §V

Three parameters, *one shape*

$$\tilde{\varepsilon} = 3\tilde{p} + 4 + \frac{3\lambda^2}{4\pi^2}\left[1 + \mathrm{sgn}(\lambda)\sqrt{\left(1 + \frac{16\pi^2}{3\lambda^2}(\tilde{p}+1)\right)}\right] \qquad \lambda \equiv \frac{a_2}{\sqrt{(a_4 B)}}$$

$p = B\,\tilde{p}$ and $\varepsilon = B\,\tilde{\varepsilon}$ · $\lambda = 0$ is the bag model · $\lambda < 0$ ($a_2 < 0$) gives $c_s^2 > 1/3$

The dimensionless EOS knows only λ, and the structure equations have no other scale. At fixed λ, masses and radii scale as $\mathbf{B^{-1/2}}$; Λ, $\bar{I}$ and the compactness depend only on λ and p_c/B.

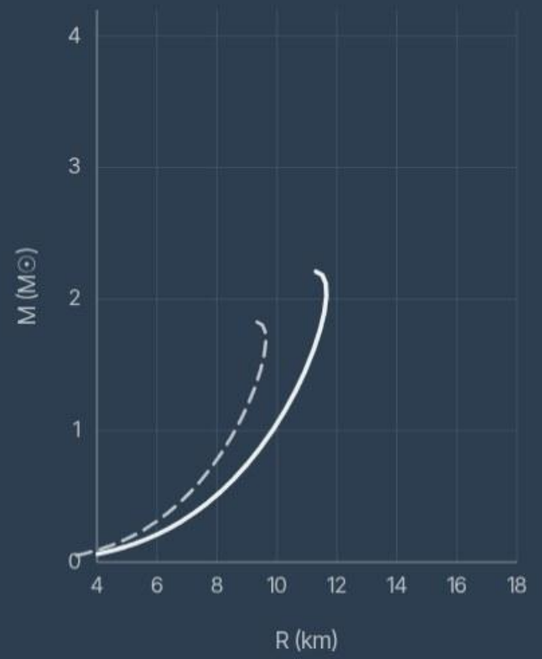
Two consequences: the observables measure only **(λ, B)**, so a_4 at fixed (λ, B) is informed only by the window; and the forward model becomes a **one-parameter** family, tabulated once.

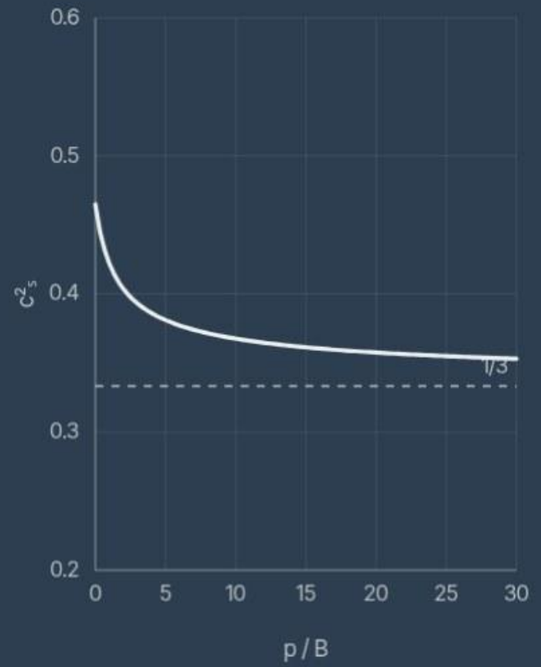
Manuscript · §II · equivalent rescaling in Zhang & Mann 2021, here for both signs of a_2

λ sets the shape, B *sets the scale*

Two stars with the same λ and different B

physical | rescaled





λ −3.4

$B^{1/4}$ (MeV) 162.0

M_{max}: 2.210 and 1.827 M⊙ · ratio 1.21000 vs $(B_2/B_1)^{1/2}$ = 1.21000

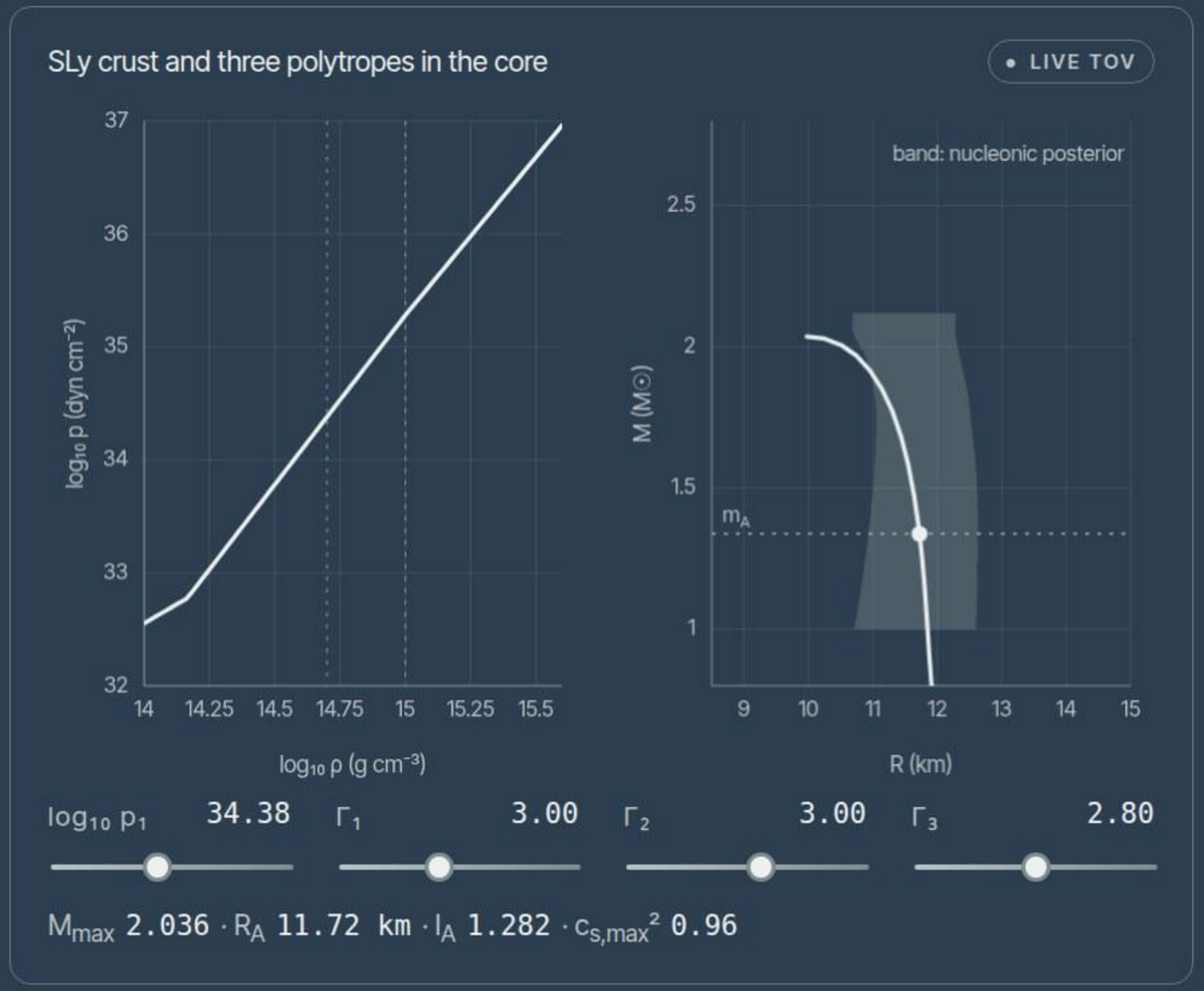
Every point is a TOV solved now, with physical B, without using the rescaling.

In rescaled mode the two curves fall on top of each other.

On the right, $c_s^2(\tilde{p})$: λ < 0 stays above 1/3. The curve depends only on λ; moving B does not change it.

Manuscript · §II · in this panel, the paper's equation of state solved in the browser

Piecewise polytropes, *no composition assumed*



Four parameters, dividing densities at $10^{14.7}$ and 10^{15} g cm^{-3}. The sliders span the N1 prior limits and start close to SLy.

We exclude EOSs that become acausal below the center of the maximum-mass star. **35.3%** of the box volume survives, and that fraction enters the evidence.

It fixes the low-density physics, with no explicit chiral effective field theory constraints and no phase transitions.

Read et al. 2009 · Carney et al. 2018 · Manuscript · §II

A forward model of *1241 sequences*

STRANGE STARS: A TABLE IN Λ

1241 values of $\lambda \in [-50, 12]$, stable branch up to the first maximum of $M(p_c)$, interpolated in λ and in $s = \sqrt{(1 - M/M_{max})}$. About 3 ms per likelihood.

NUCLEONIC STARS: DIRECT SOLUTION

About 5 s per admissible EOS, hence the prior library of Part III. The SLy polytrope fit gives 2.048 $M_\odot$, 11.74 km, $\Lambda = 398$ and 1.285×10^{45} g cm^2, against 2.048, 11.73, 400 and 1.287 for the tabulated SLy4.

Table against 4000 direct solutions

QUANTITY	LARGEST RELATIVE ERROR
R	6×10^{-6}
Λ	2×10^{-6}
I	5×10^{-7}
M_{max}	10^{-7}

Half of the points within 3% of M_{max}. The solutions use the original (a_2, a_4, B), not the rescaled EOS, so they also verify the homology.

Manuscript · §III and §VI · TOV, Hartle 1967, Hartle & Thorne 1968, Love number with surface correction (Damour & Nagar 2009; Postnikov et al. 2010)

III

PART III

The inference

Public data, the same likelihoods and samplers that watch each other



MAIN ANALYSIS

Six pieces of *public data*

OBJECT	MEASUREMENT	SOURCE	USE
PSR J0348+0432	$M = 2.01 \pm 0.04\ M_\odot$	radio, Antoniadis 2013	Gaussian in M
PSR J1614−2230	$M = 1.937 \pm 0.014\ M_\odot$	radio, NANOGrav 15-yr	Gaussian in M
PSR J0740+6620	(M, R_{eq}) samples	NICER×XMM, Salmi 2024	KDE, includes radio mass
PSR J0437−4715	(M, R_{eq}) samples	NICER, Choudhury 2024	KDE, includes radio mass
GW170817	$(m_1, m_2, \Lambda_1, \Lambda_2)$ samples	GWTC-1, low spin	KDE in $(q, \Lambda_1, \Lambda_2)$
PSR J0737−3039A	$m_A = 1.338183\ M_\odot$	timing, Kramer 2021	conditioning

> J0030+0451 is left out of the main analysis: its updated analyses give different, multimodal solutions. It enters as a variant, as does the NICER analysis of J1614.

Manuscript · Table I · files and checksums in the supplementary material

LIKELIHOOD

One population, *marginalized masses*

$$\mathscr{L}_k(\theta) = \int dM\, p(\, d_k \mid M, X(M;\theta)\,)\, \pi(M \mid \theta) \qquad \pi(M \mid \theta) = \frac{1}{M_{max}(\theta) - M_{low}}$$

X is the radius or the deformability given by the EOS · $M_{low} = 1\, M\odot$

The normalization penalizes EOSs whose maximum mass greatly exceeds the observed masses. That is a choice of population model, not a datum.

So there is a variant with an EOS-independent density, $1/(M_{up} - M_{low})$ on $[M_{low}, M_{max}]$, with $M_{up} = 3.5\, M\odot$. In it, ln $\mathscr{B}$ goes from +0.21 to +0.98.

Manuscript · §IV and §X

NICER: the density *already contains the radio mass*

The priors of the X-PSI analyses are uniform in R_{eq} on $[3r_g(1\ M\odot), 16\ km]$ and already contain the radio mass.

So the posterior density in (M, R_{eq}) is the X-ray likelihood times the radio measurement, divided by a constant. We use a KDE of it as $p(d \mid M, R)$ and **do not add those masses again**.

DO THE PRIOR CUTS MATTER?

Where the prior vanishes the estimate vanishes, which acts as an exclusion. For J0740 the binding cut is the compactness one, $3GM/c^2 = 9.2$ km at $2.07\ M\odot$.

It lies below the 0.1% points of $R_{2.0}$ in both posteriors: 10.4 and 10.1 km. No posterior EOS approaches 16 km.

Salmi et al. 2024 · Choudhury et al. 2024 · Fonseca et al. 2021 · Manuscript · §IV

TIDAL DEFORMABILITY

GW170817 *without hadronic relations*

$$\mathscr{L}_{\mathrm{GW}}(\theta) \propto \int \mathrm{d}q\, \hat{p}(q, \Lambda(m_1;\theta), \Lambda(m_2;\theta) \mid d)\, \pi(m_1, m_2 \mid \theta)$$

Masses at the measured chirp mass, $\mathscr{M}_c$ = 1.1858 M⊙, known to 6×10^{-5} · $\pi(m_1, m_2 \mid \theta) = 2/(M_{max} - M_{low})^2$

WE USE

GWTC-1 low-spin samples, with Λ_1 and Λ_2 independent and uniform on [0, 5000]. Nothing in them assumes an EOS.

WE DO NOT USE

The samples imposing a common EOS through relations calibrated on hadronic EOSs, nor the maximum-mass bound inferred from the remnant, which assumes hadronic matter.

Abbott et al. 2019 · Abbott et al. 2018 · Manuscript · §IV

MOMENT OF INERTIA OF PULSAR A

Explicit priors, *with variants*

Strange stars

P1	uniform in $(a_2, a_4, B^{1/4})$, $0.05 \le a_4 \le 1$, $-3\alpha \le a_2 \le \alpha a_4$, B in the window
P2	uniform in B
P3	$a_2 \ge -6\alpha$
P4	$a_4 \ge 0.5$

Nucleonic stars

N1	$\log_{10} p_1 \in [33.6, 35.4]$, $\Gamma_1 \in [2, 4.5]$, $\Gamma_{2,3} \in [1.1, 4.5]$, $c_s \le 1$
N2	tolerance $c_s \le 1.1$
N3	wider box, $\log_{10} p_1 \in [33.4, 35.6]$, $\Gamma_i \in [1.1, 5]$

Carney et al. 2018 · Manuscript · §II and §V

SAMPLERS

Each family, *two independent paths*

	STRANGE STARS	NUCLEONIC STARS
Posterior	**emcee**, DE moves only, 4 ensembles, ≥ 436 τ, ~91×10^3 effective samples	**Prior library**, 6×10^5 weighted EOSs, 3944 effective samples
Cross-check	dynesty, 3 runs: percentiles within 2.4% of the 90% width	emcee (1%) and dynesty (5%) with the direct forward model
ln Z	prior average: **−19.37 ± 0.005**	library: **−19.58 ± 0.02**
ln Z, check	dynesty: −19.33 (scatter 0.04)	dynesty: −19.59 ± 0.06

> The dynesty error for nucleonic stars comes mostly from counting the volume with nonzero likelihood: only 10.8% of the box.

Manuscript · §V and §VI

A BIASED MOVE

Every diagnostic passed, and *the tail was wrong*

The first nucleonic emcee run mixed DE (80%) and DE-snooker (20%). It passed every convergence test, but put **0.138** and **0.039** of the mass above the library's 84 and 95% points of I_A, instead of 0.16 and 0.05.

That is 4.8 and 5.7 standard errors. Nested sampling agreed with the library.

```
# emcee 3.1.6, DESnookerMove
u = delta / sqrt(norm)
q = s + u * gamma * (dot(u, z1) - dot(u, z2))
return q, 0.5 * (ndim - 1.0) * metropolis
```

The step grows with the distance to the reference walker, and the Jacobian factor has half the exponent of ter Braak & Vrugt (2008). The move does not leave the target invariant.

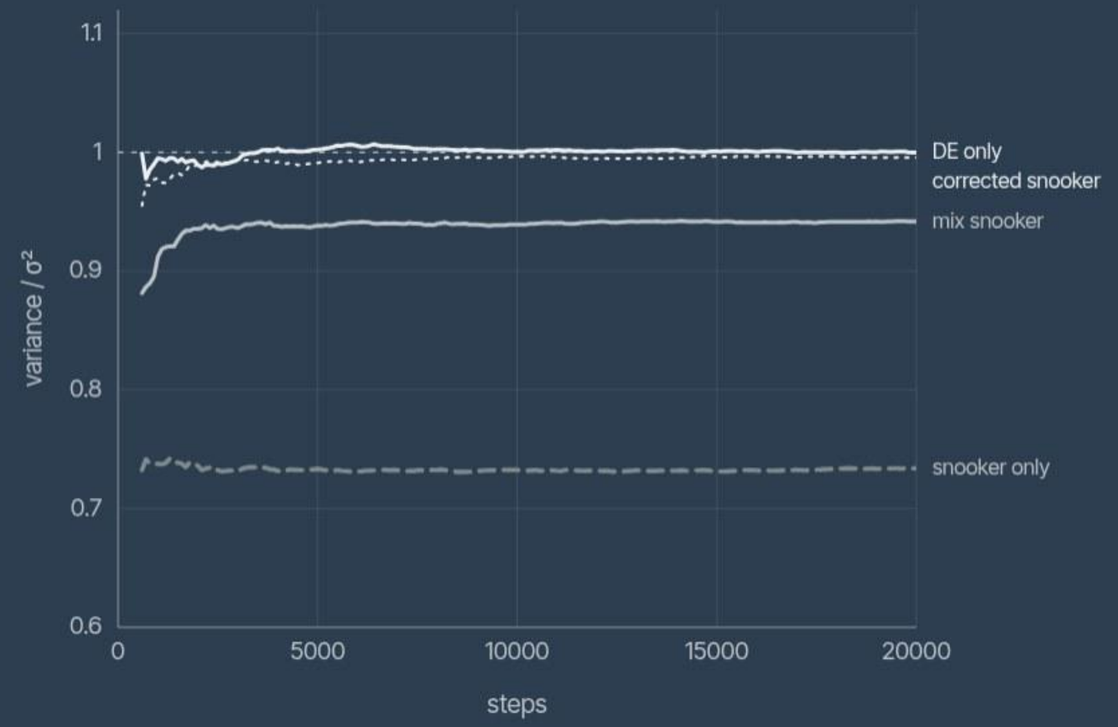
Manuscript · §VI · ter Braak & Vrugt 2008 · emcee/moves/de_snooker.py, version 3.1.6

LIVE

The test that *exposes the bias*

64 walkers, standard normal in 4 dimensions, scaled by σ

σ = 0.1 | σ = 1 | σ = 100



Step `20000` · DE `1.000` · mix `0.942` · snooker `0.734` · corrected `0.996` · snooker acceptance `26.0%`

The same emcee moves run now in the browser. A correct target gives variance 1.

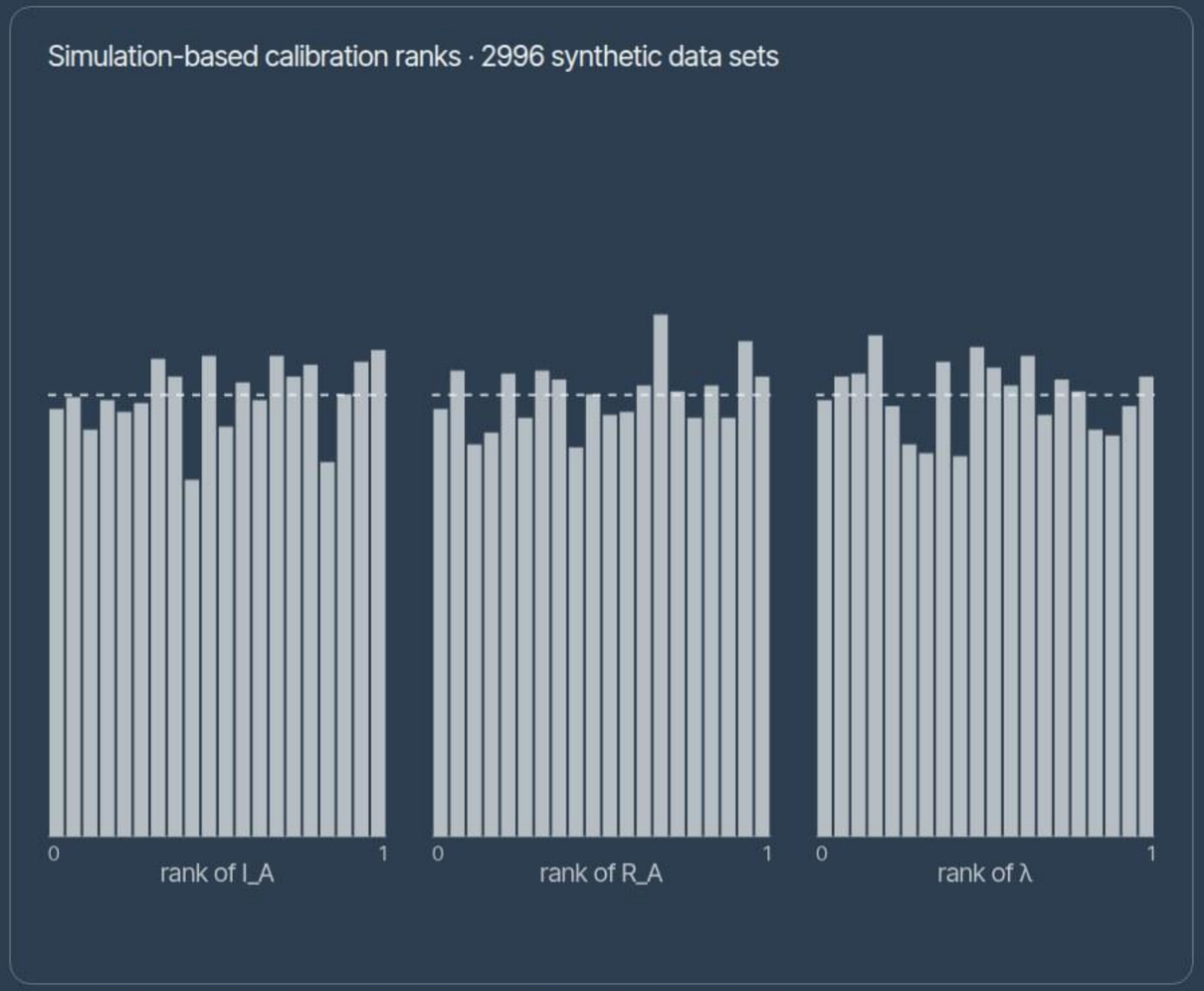
With 20 thousand steps: **0.73** with snooker alone, **0.94** in the 80/20 mixture and **1.002** with DE alone.

The bias depends on the units. The step scales with the parameters: for width 0.1 the move is accepted for 91% of the proposals, and for width 100 for 0.2%.

In (a_2 [MeV^2], $B^{1/4}$ [MeV]) it was inert: 0 of 7200 proposals accepted. In the nucleonic parameters, it was not.

Manuscript · §VI · in this panel, a JavaScript reimplementation of DEMove and DESnookerMove

Calibrated, *given the likelihoods*



Manuscript · §VI · a flat histogram (dashed) is what a calibrated chain gives

89.7%

coverage of the 90% interval of I_A; 90.0% for R_A. Kolmogorov–Smirnov p = 0.49

Synthetic data with the structure of the real data: two radio masses, two (M, R) measurements and a GW170817-like binary.

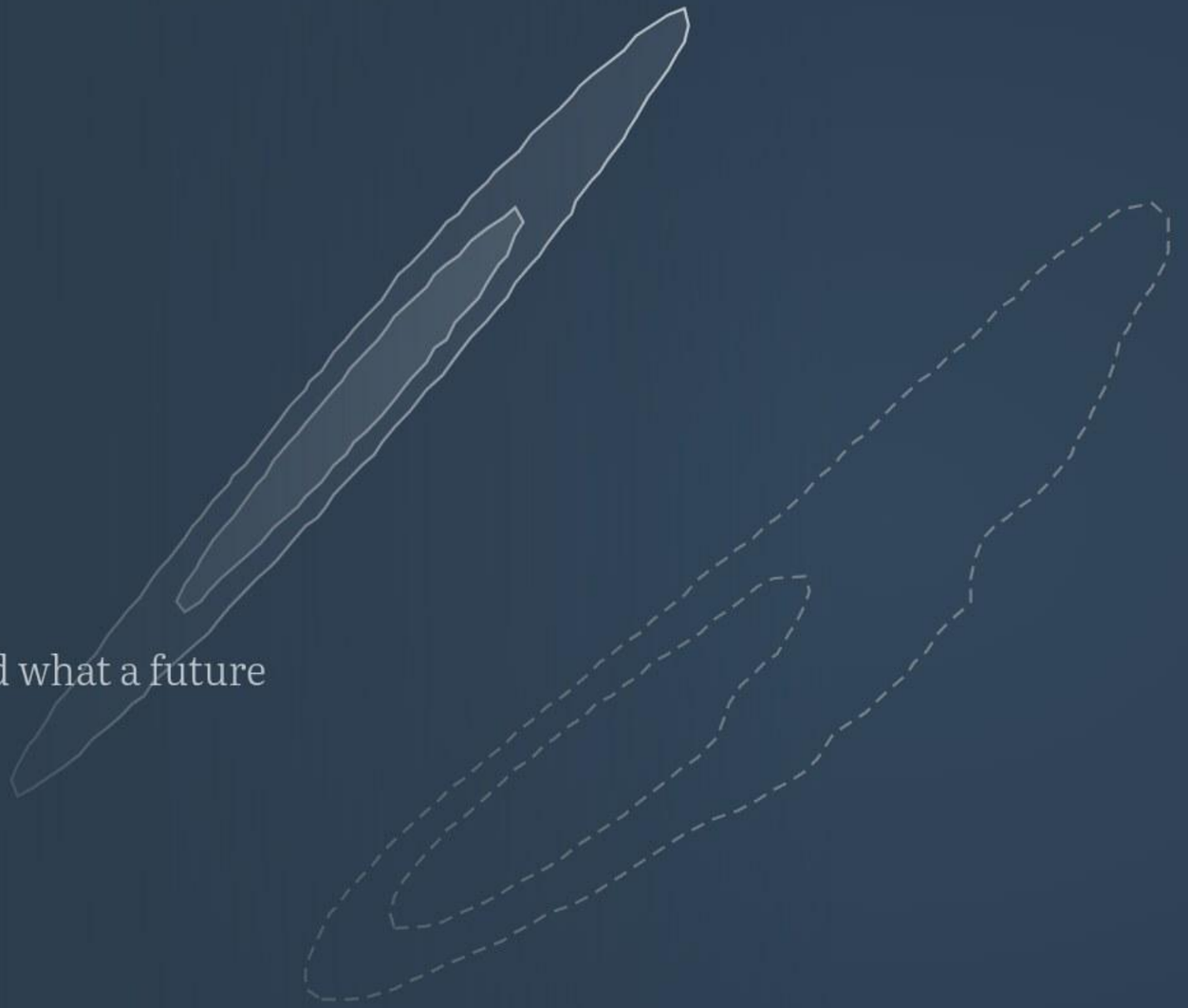
> This tests the inference **given likelihoods of the assumed form**. It does not test their construction from NICER and GW170817, nor the applicability of those analyses to strange stars.

IV

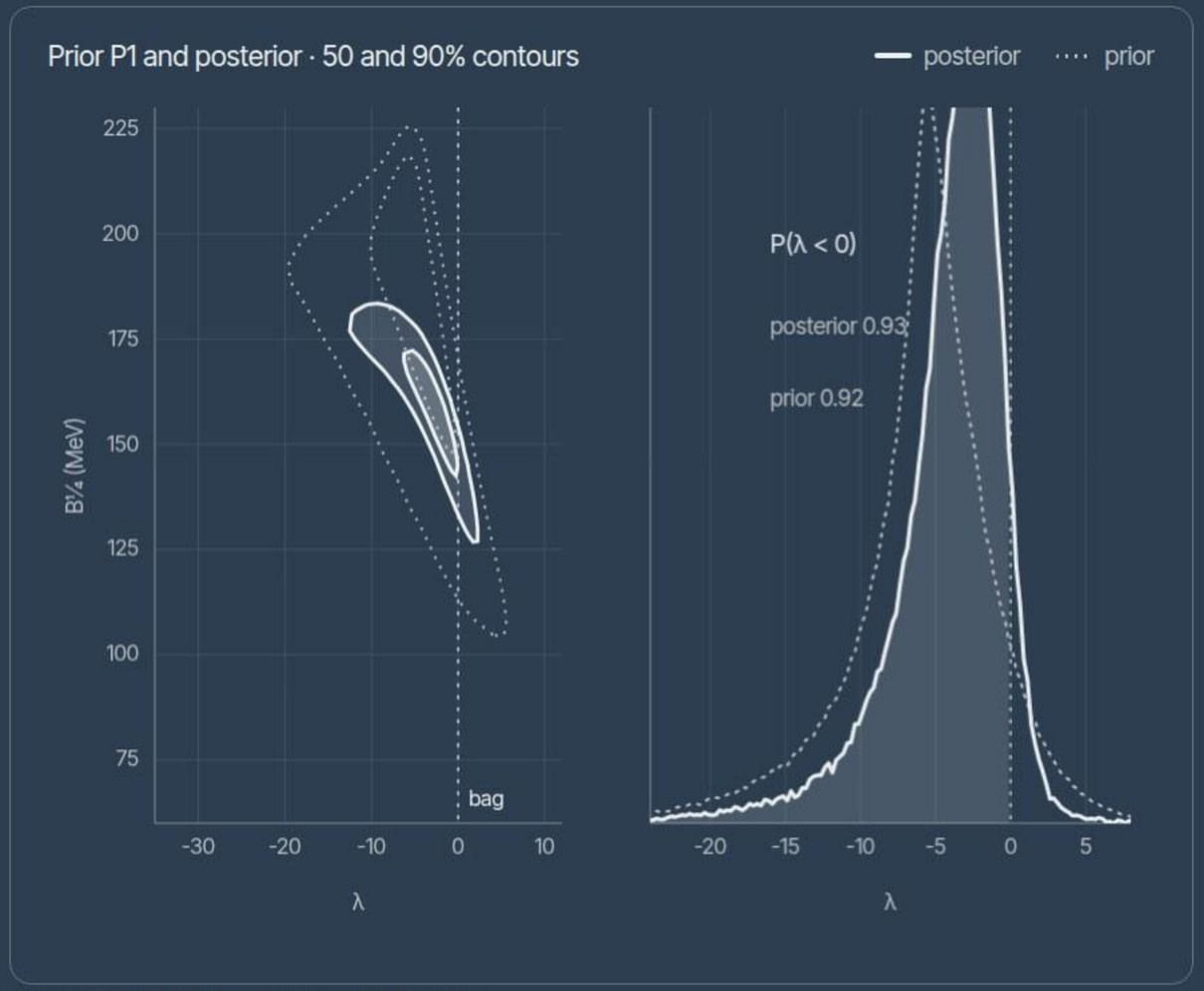
PART IV

Results

Narrow predictions, evidence of order unity and what a future measurement would change



The data narrow λ, *not its sign*



Manuscript · §VII · M_{max} = 2.21 (2.04–2.61) M⊙

$$\lambda = -3.4$$

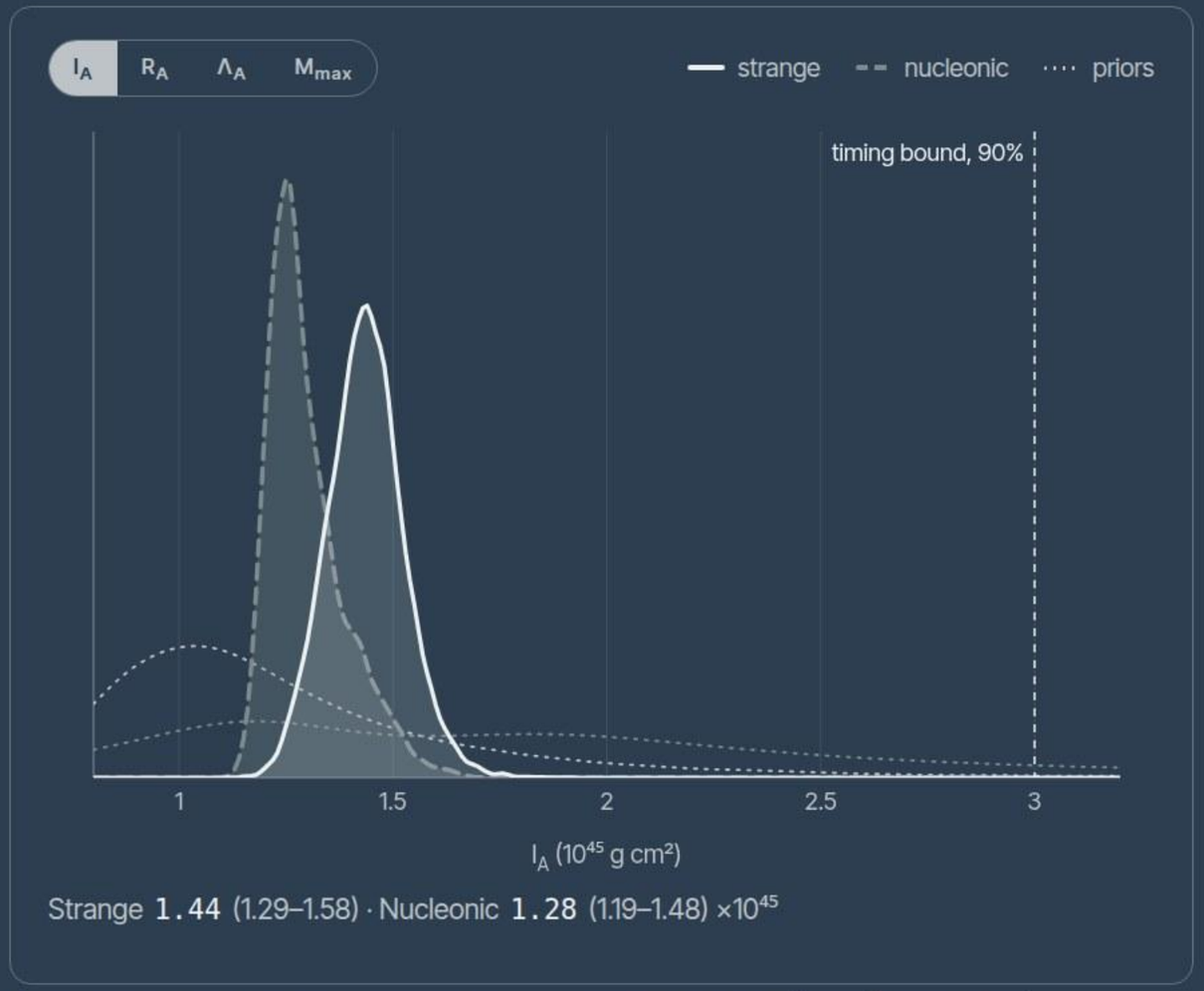
−11.4 to +0.4 (90%), against −15 to +1 in the prior

P(λ < 0), posterior	0.93
P(λ < 0), prior	0.92
Odds ratio	1.1

The preference for $a_2 < 0$ comes from the prior domain, which extends to −3α below zero but only to αa_4 above it. The data do not say whether the matter is stiffer than the bag model.

PREDICTIVE DISTRIBUTIONS AT THE MASS OF A

The prediction for *pulsar A*

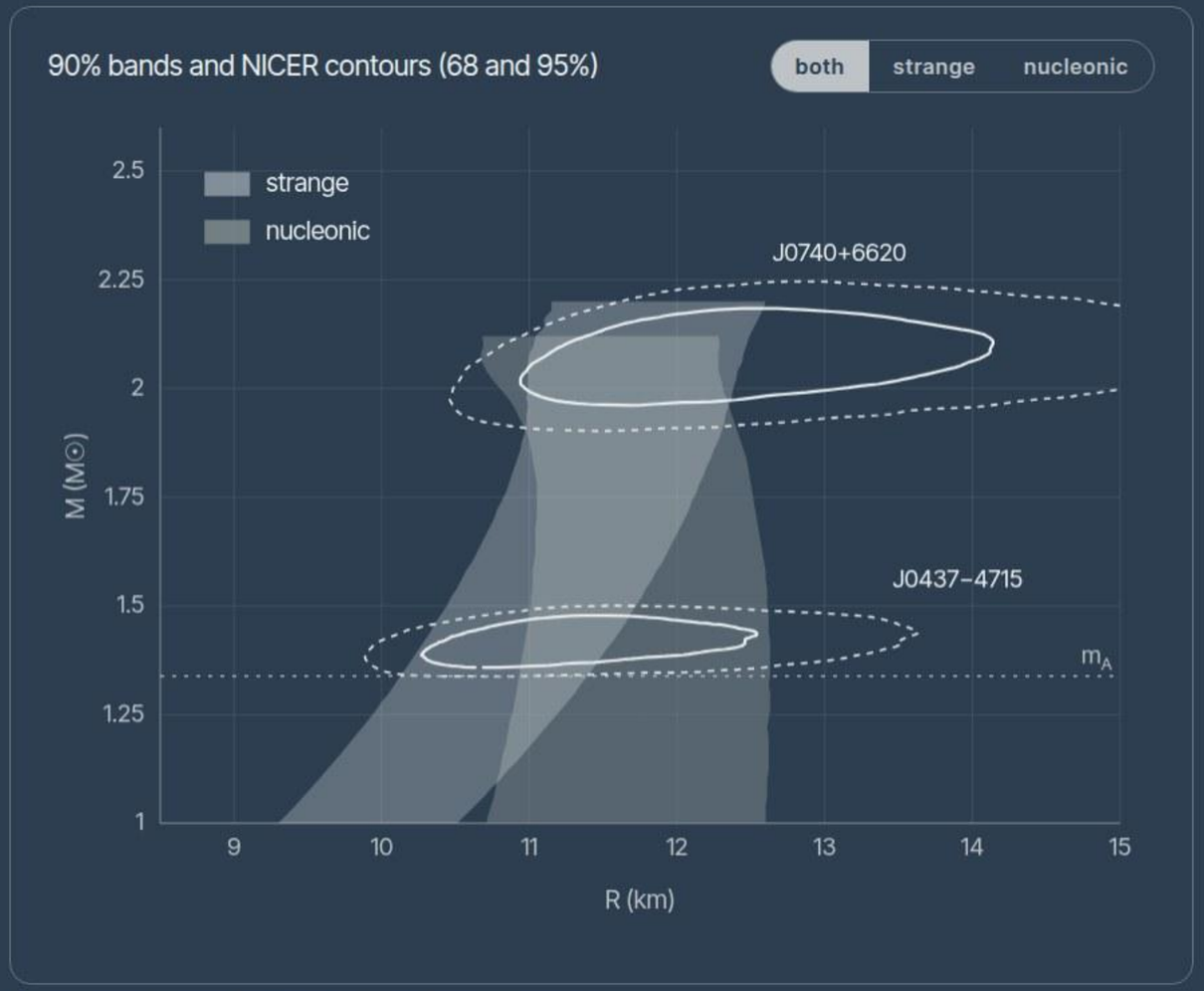


	STRANGE	NUCLEONIC
I_A (10^{45})	1.44	1.28
	1.29–1.58	1.19–1.48
R_A (km)	10.8	11.5
	10.1–11.4	10.9–12.6
Λ_A	599	385
	404–860	292–669

The data reduce the width of the strange-star I_A interval about fivefold. The timing bound carries no information within this family.

Manuscript · §VII and §VIII · medians and 90% intervals · prior: 0.78–2.26 × 10^{45} g cm² (strange stars)

Two bands *against NICER*



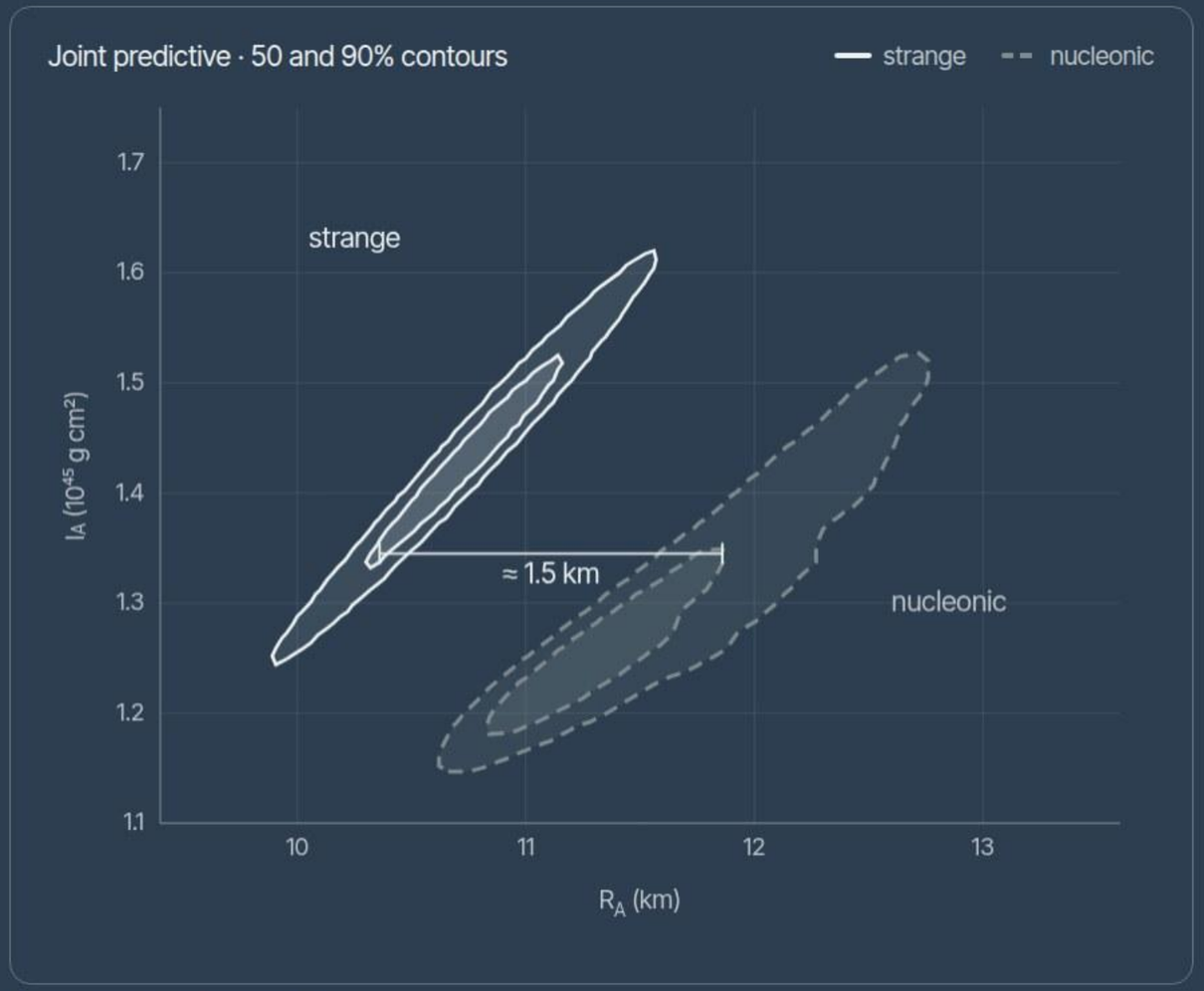
	STRANGE	NUCLEONIC
M_{max} (M⊙)	2.21	2.14
$R_{1.4}$ (km)	10.9	11.5
$\Lambda_{1.4}$	477	296
$R_{2.0}$ (km)	11.7	11.4

The curves cross: at 1.4 M⊙ strange stars are smaller; at 2.0 M⊙, slightly larger. The strange maximum mass reaches further: 95% point at 2.61 against 2.31 M⊙.

Manuscript · Table II · Salmi et al. 2024 · Choudhury et al. 2024

THE (I, R) PLANE OF PULSAR A

Overlapping in I, *separated in the plane*



≈ 1.5 km

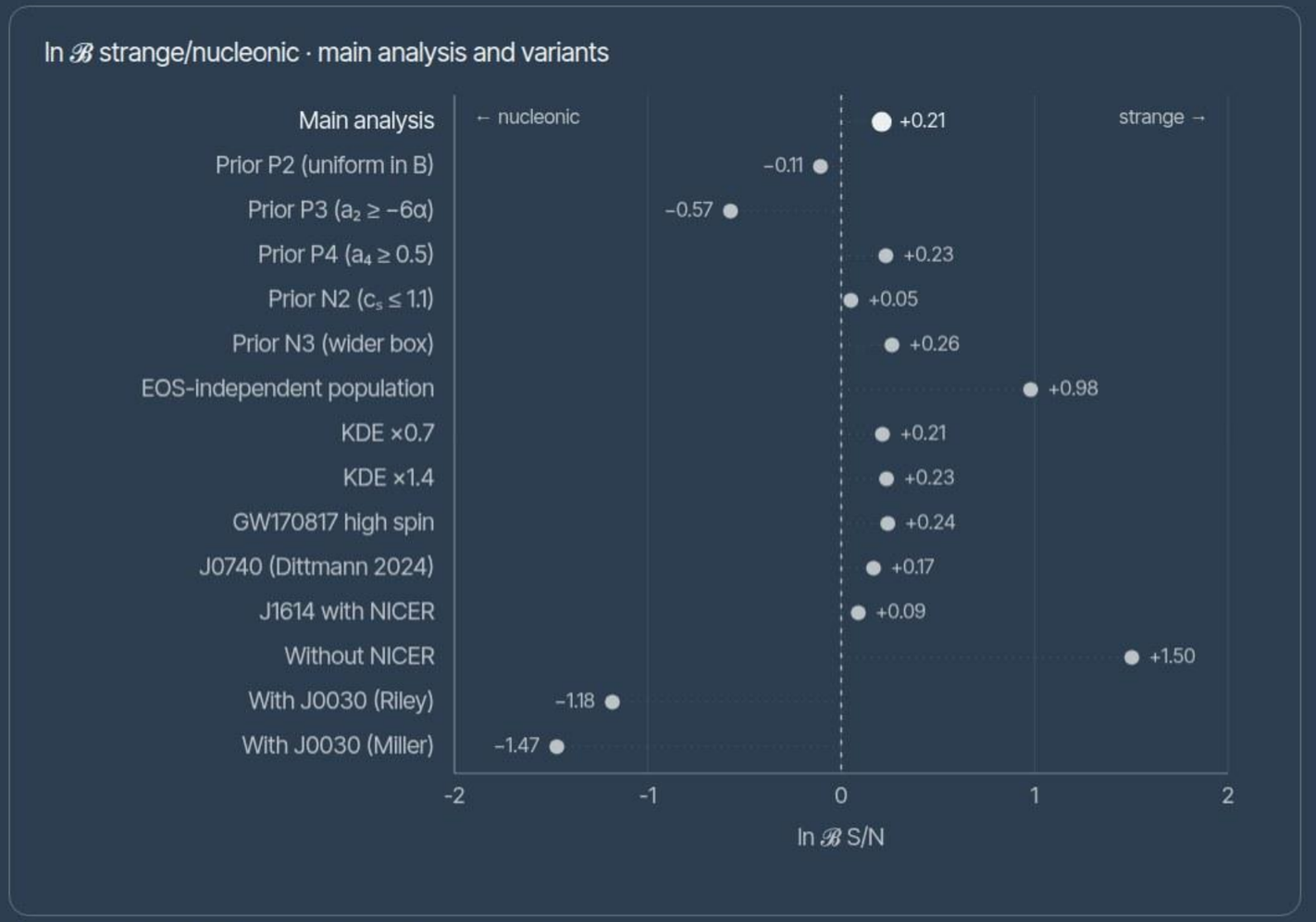
radius difference between the families at the same I_A

Nearly uniform density gives more moment of inertia per radius. So the same measurement of I_A points to different radii in each family.

I_A and Λ_A follow nearly the same relation in both families: the separation is in the radius.

Manuscript · §VIII · appendix A

$\ln \mathcal{B} = +0.21$, and it *changes sign*



Manuscript · §VIII · evidence table · Monte Carlo errors ~0.02

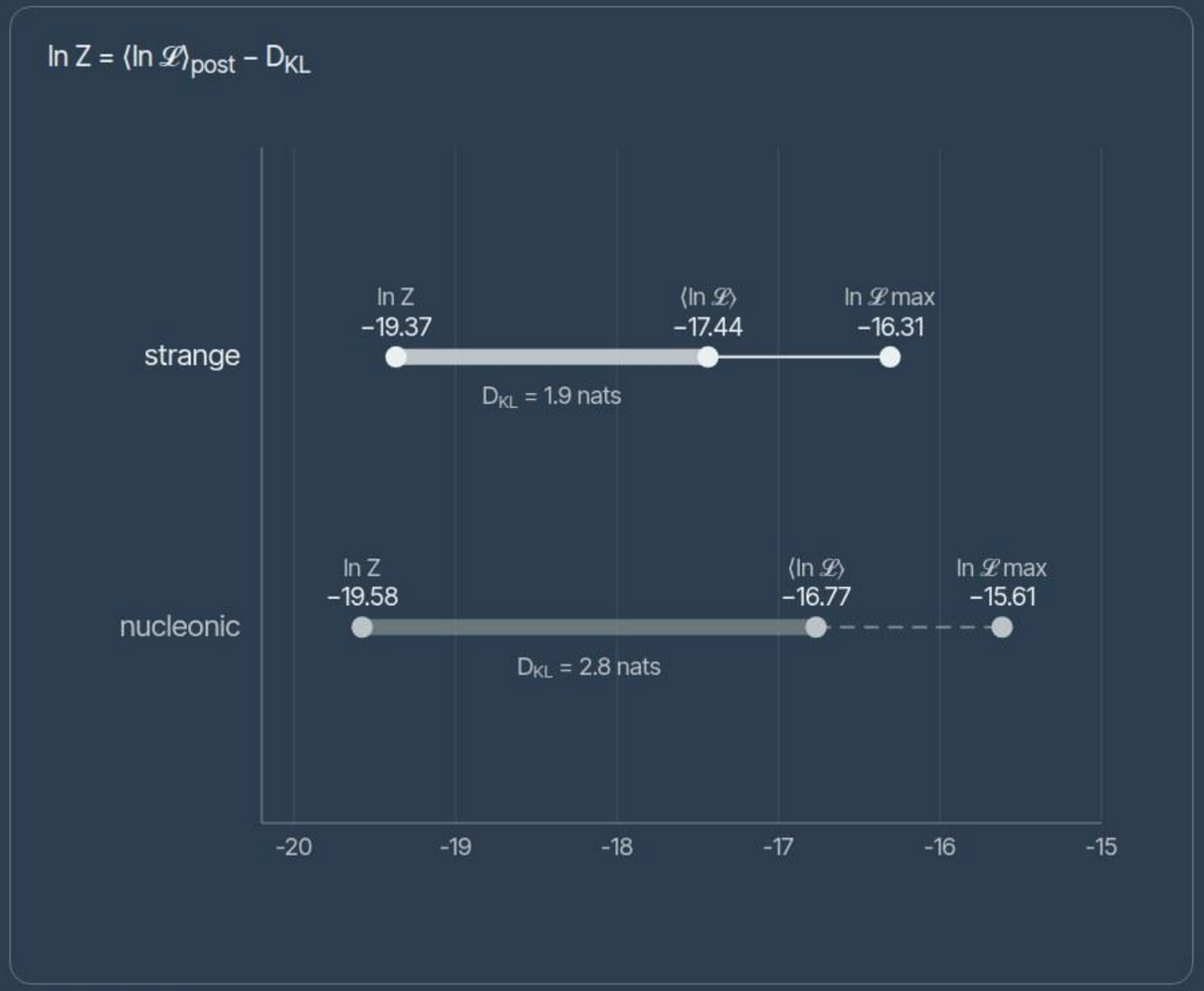
−1.5 to +1.5

range across all prior and data variants

Without NICER, strange stars are mildly preferred. With J0030, in either analysis, the preference goes to nucleonic stars.

Changing priors within each family moves ln $\mathcal{B}$ by up to about 0.8, the size of the preference itself.

The sign comes from volume, *not from fit*



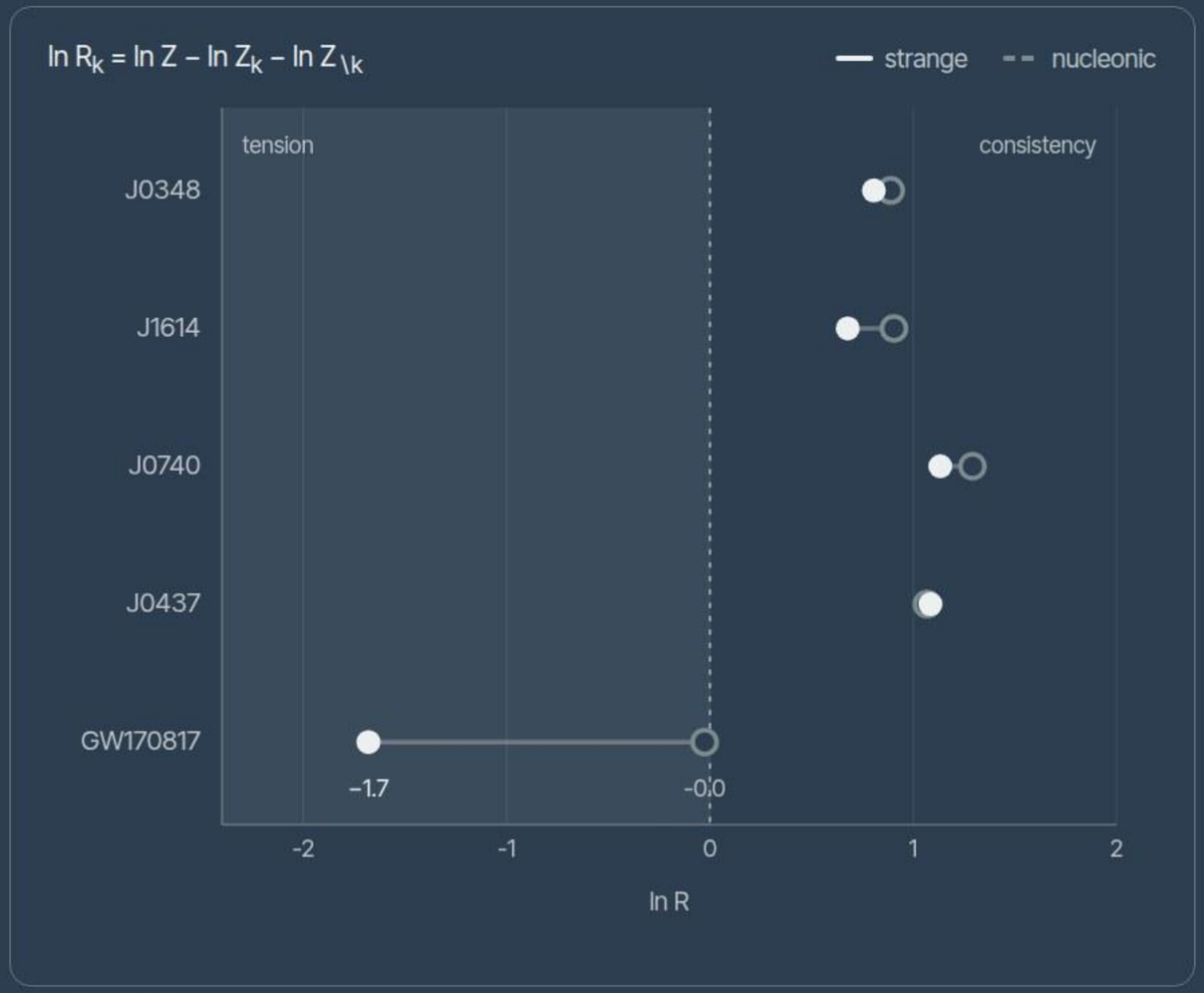
Nucleonic stars fit better on average and at the best point: $\langle \ln \mathscr{L} \rangle = -16.77$ against -17.44; maxima -15.61 and -16.31.

But their posterior occupies a smaller fraction of their prior: $D_{KL} = 2.8$ against 1.9 nats.

> The small positive $\ln \mathscr{B}$ comes from the **concentration within the prior**, not from a better description of the data.

Manuscript · §VIII

GW170817 pulls strange stars *the other way*



Positive: the data set is more probable given the others than a priori. The radio masses and the two NICER pulsars are consistent: ln R between +0.7 and +1.3.

For nucleonic stars GW170817 is neutral, ln R = 0.0. For strange stars there is **mild tension**, ln R = −1.7: GW170817 alone prefers softer stars than the maximum mass and J0740 require.

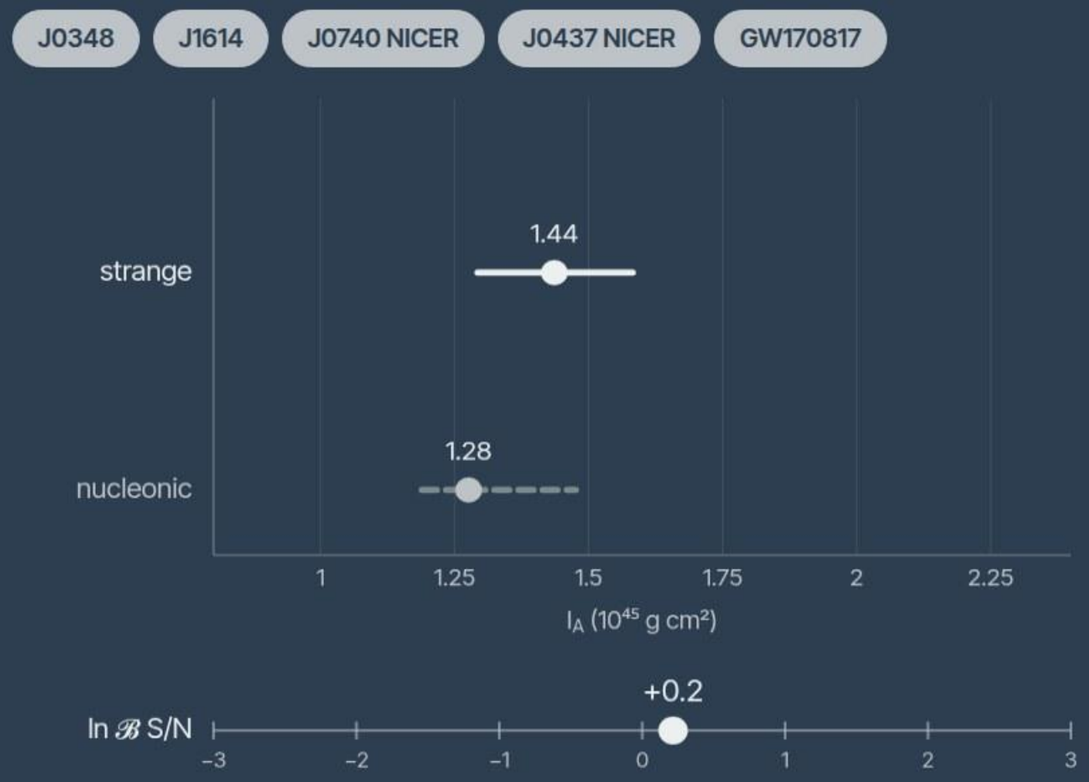
Alone it would give ln $\mathcal{B}$ = +0.8; without it, +1.1. The tension brings the combined value close to zero.

Manuscript · §VIII

MOMENT OF INERTIA OF PULSAR A

EVIDENCE OF EACH SUBSET

Switch the data *on and off*



ln $\mathscr{B}$ `+0.2` · strange I_A `1.29–1.58`, R_A `10.8 km` · nucleonic I_A `1.19–1.48`, R_A `11.5 km`

The 48 combinations use the paper's construction: prior average over 2×10^5 draws and the N1 library.

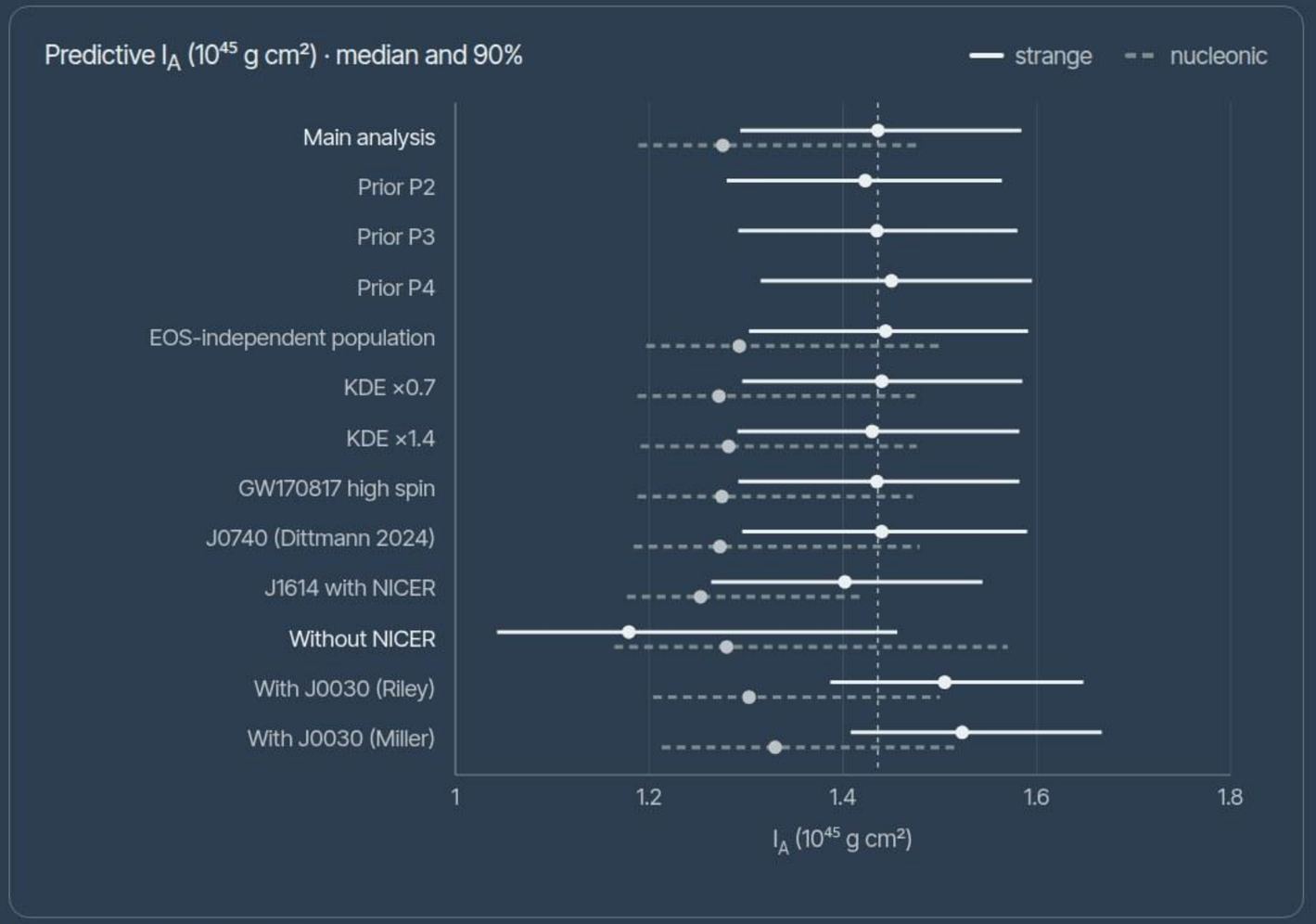
J0740 has three states, because the NICER density already contains the radio mass. With J0740 radio only and J0437 off you get the no-NICER variant: strange I_A drops to 1.18.

The larger I_A of strange stars is a consequence of the NICER radii, not of the masses or of GW170817.

Computed for this seminar with the analysis code (tools/subconjuntos.py) · ln $\mathscr{B}$ to one decimal: the draw is that of the consistency analysis of §VIII

SENSITIVITY

The prediction rests *on NICER*



Prior variants move the strange median only between 1.42 and 1.45.

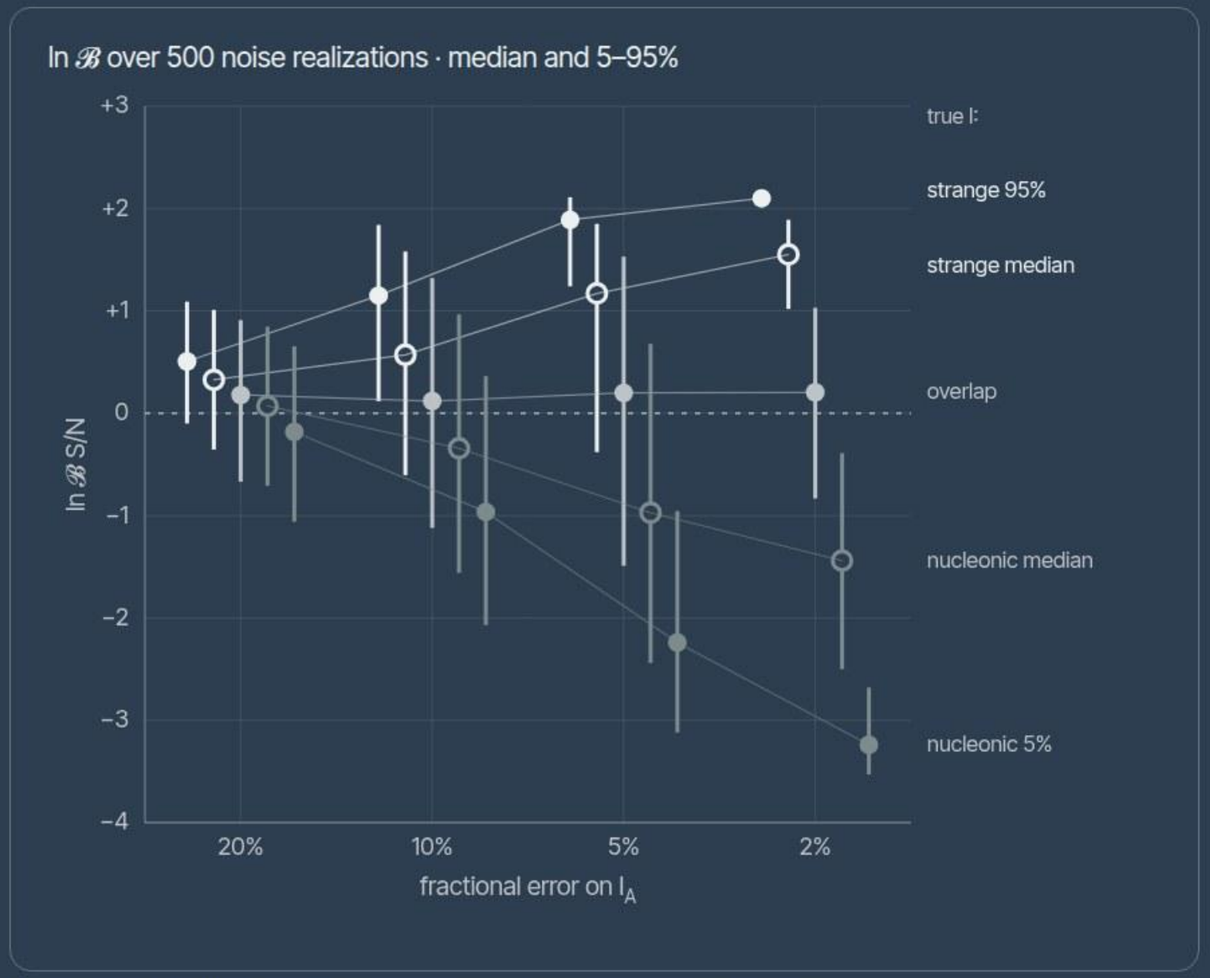
Data variants that keep NICER: 1.40 to 1.52.

Without NICER: 1.18 (1.04–1.46).

J0030 has a larger radius and raises I_A; the NICER analysis of J1614 has a smaller radius and lowers it.

Manuscript · §X · variants table

I_A alone separates little *in the simulated cases*



In the overlap (I_A = 1.35), even at 2%: median **+0.2**, with 90% of the realizations between −0.8 and +1.0.

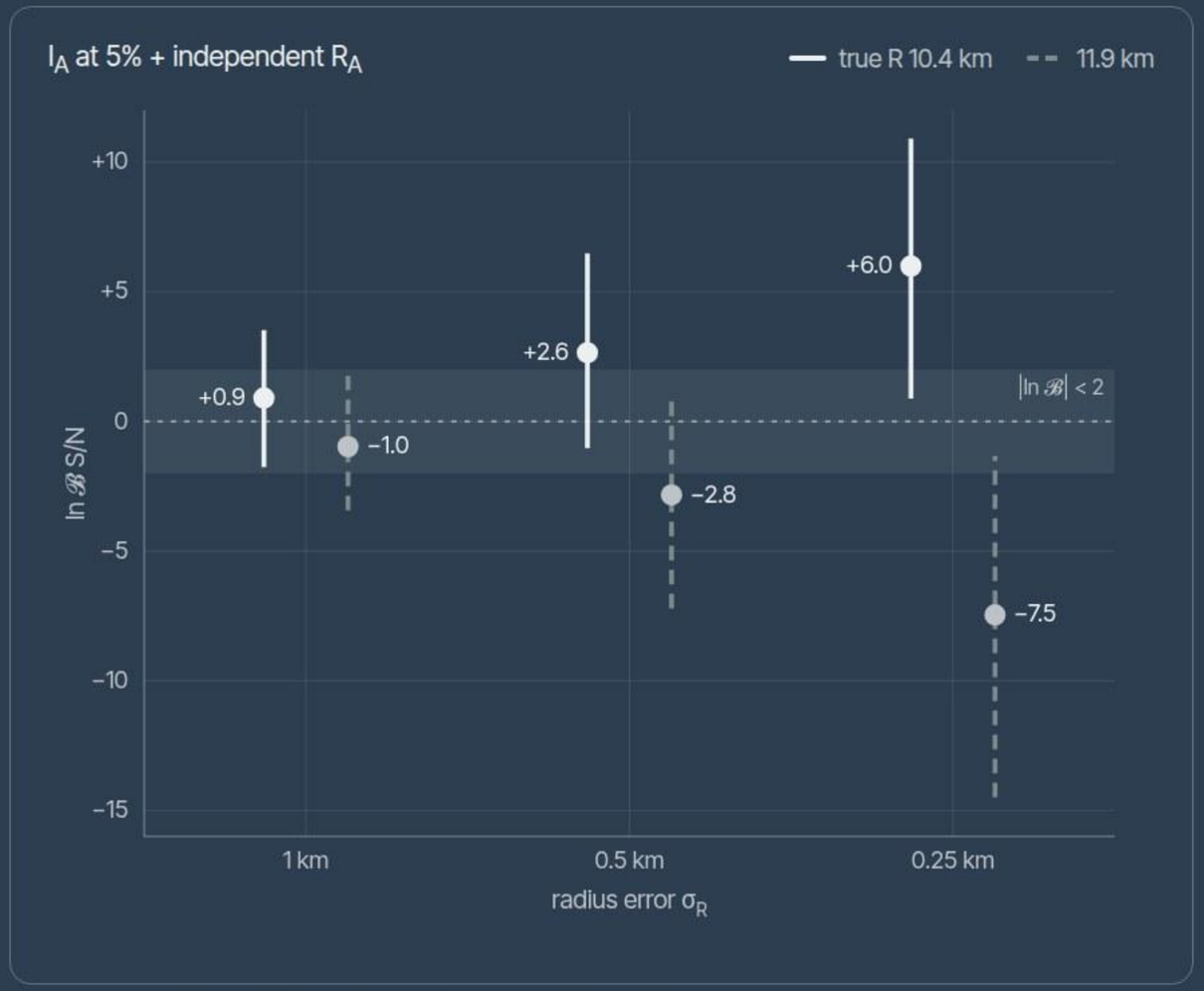
From the nucleonic 5% point to the strange 95% point, the median $|\ln \mathcal{B}|$ at 2% is at most **3.2**: odds of about 25:1, for equal prior odds.

At 10%, the precision anticipated around 2030, the median stays below 1.2 in all cases.

> Statements about the simulated cases. True values beyond these points, improbable under both families, would give larger $|\ln \mathcal{B}|$.

Manuscript · §IX · illustrations of information content, not instrumental forecasts

The radius acts *where the families differ*



Σ_R	$\lvert\ln \mathcal{B}\rvert > 2$, RIGHT SIGN	WRONG SIGN
1 km	24%, 24%	2.8%, 3.4%
0.5 km	62%, 64%	1.6%, 2.6%
0.25 km	91%, 91%	0.2%, 0.4%

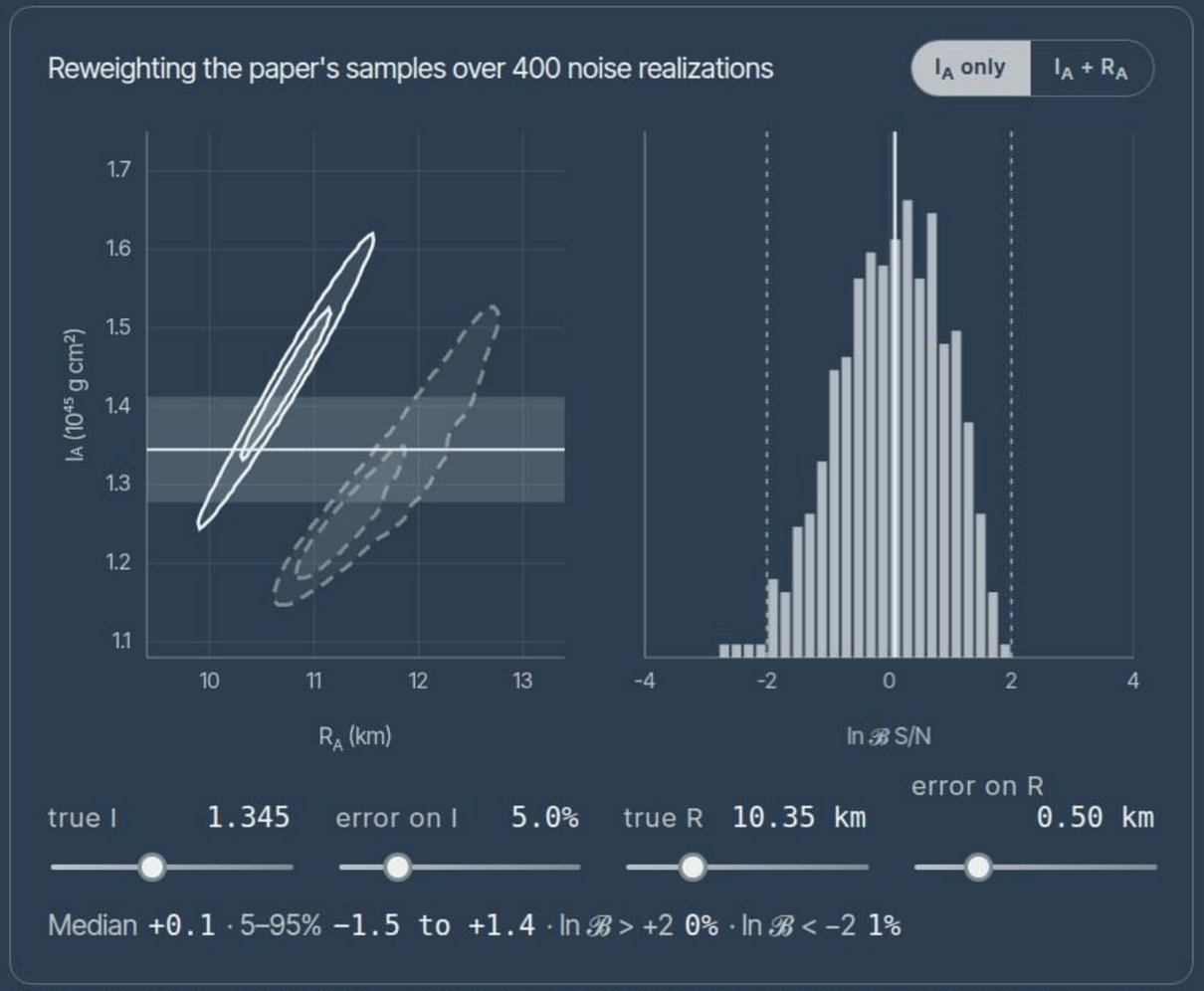
Bare strange stars. A nuclear crust would add a few hundred meters, the size of these errors.

No independent radius measurement of pulsar A **exists or is planned**. Radii of other stars inform R_A only under a shared EOS, and a radius inferred from I_A through a universal relation would not be independent.

Manuscript · §IX · pairs of values: true radius strange, nucleonic

LIVE

Simulate *a measurement*



Manuscript · §IX · (I_A, R_A) grid from 2×10^7 draws (strange stars) and the N1 library · see tools/validacao_grade.txt

Each realization adds noise to the measurement and updates both evidences, as in the paper:

$$Z' = Z \cdot E_{post}[\, N(d; x, \sigma)\,]$$

The panel uses the same samples on a fine grid: it reproduces the paper's projections within the noise of 500 realizations.

Try I = 1.19 at 2%, or the strange radius against the nucleonic one at 0.5 km.

LIMITATIONS

What is *not in the model*

1 Crust and atmosphere

Bare strange stars; a thin crust would add hundreds of meters. NICER assumes a hydrogen atmosphere and shapes calibrated on hadronic stars, which amounts to a thin crust.

2 Rotation

Non-rotating forward model. With second-order Hartle on 300 EOSs: J0740 gains up to 0.22 km (strange) and 0.32 km (nucleonic); J0437, up to 0.08 km, far below the NICER uncertainties. M_{max} rises by up to 0.022 $M_\odot$.

3 The families themselves

No mixed binaries, no hybrid family, no phase transitions, no explicit chiral constraints. Discrimination refers to these two parameterizations.

Manuscript · §IV and §X

CONCLUSIONS

A narrow prediction, *an open evidence*

1

$I_A = 1.44\ (1.29–1.58)$, $R_A = 10.8\ (10.1–11.4)$ km

Robust to the priors, resting on the NICER radii. The sign of λ stays where the prior put it.

2

Current data do not discriminate these families

$\ln \mathcal{B}$ of order unity, reflecting the concentration within the priors, not the quality of the fit.

3

GW170817 in mild tension within the strange family

With the maximum mass and the NICER radii. Not within the nucleonic family.

4

The information is in the (I_A, R_A) plane

I_A alone at 2%: median $|\ln \mathcal{B}|$ up to 3.2 in the simulated cases. A hypothetical independent radius with a 0.5 km error, for bare strange stars, would give $|\ln \mathcal{B}| > 2$ with the right sign in 62–64% in the overlap; no such measurement exists.

Conclusions conditional on the two parameterizations, the data selection and the assumptions of §X · Manuscript in preparation

SOURCES AND CREDITS

References and *data*

This work
R. C. R. de Lima, J. V. Zastrow, J. P. Pereira, Bayesian predictions for the moment of inertia of PSR J0737−3039A and tests of the strange-star hypothesis. Manuscript in preparation, authorship to be confirmed.

Data
Salmi et al. 2024 · Choudhury et al. 2024 · Abbott et al., PRX 9 (2019) 011001 and 031040 · Antoniadis et al. 2013 · Agazie et al. 2023 · Fonseca et al. 2021

Numbers
Plots from the paper's outputs, with provenance in assets/data-sources.json. The TOV + Hartle panels and the emcee moves run in the browser.

Double pulsar
Kramer et al., PRX 11 (2021) 041050 · Burgay et al., Nature 426 (2003) 531 · Lyne et al., Science 303 (2004) 1153 · Hu et al., MNRAS 497 (2020) 3118 · Hu & Freire (2024)

Models and methods
Alford et al. 2005 · Zastrow et al., PRD 112 (2025) 083010 · Read et al. 2009 · Carney et al. 2018 · Foreman-Mackey et al. 2013 · Speagle 2020 · ter Braak & Vrugt 2008

Palette
#2C3E50, #34495E, #7F8C8D, #BDC3C7, #ECF0F1, chosen by the author. No hue: the families are told apart by light and line.

PSR J0737−3039A · BAYESIAN INFERENCE

Questions?

The moment of inertia of pulsar A and the strange-star hypothesis

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with João Vitor Zastrow and Jonas P. Pereira